Customer Shopping Behaviour Analysis

An end-to-end analysis using **Machine Learning (ML)** and **Natural Language Processing (NLP)** to uncover purchasing patterns, predict subscription decisions, and generate actionable business insights.

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Project Objectives

1

Exploratory Data Analysis

Identify patterns in customer purchasing behaviour

2

Sentiment Analysis

Analyze customer reviews using NLP techniques

3

ML Prediction

Build a model to predict subscription status

4

Business Insights

Generate actionable recommendations from findings

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Dataset & Tools

Dataset Features

Feature	Description
Age / Gender	Customer demographics
Purchase Amount	Amount spent (USD)
Previous Purchases	Number of past purchases
Subscription	Target variable (Yes/No)
Review	Customer feedback text

Dataset: **3,900 entries**, 19 columns

Tools & Libraries

Python (Google Colab)

Pandas, NumPy, Scikit-learn

Visualization

Matplotlib, Seaborn, WordCloud

NLP

NLTK, TextBlob

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Data Preprocessing

Encode Categoricals

Converted categorical fields to numeric codes.

Clean Text

Removed punctuation and stopwords from text.

Handle Missing Values

Filled missing reviews; Gender had 3,900 nulls.

Engineer Features

Created Customer Type: New, Returning, Loyal.

Key Note

The **Gender** column had 3,900 missing values — the only column with nulls. A new **Customer Type** feature was engineered:

New

0 previous purchases

Returning

1–5 purchases

Loyal

>5 purchases

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Exploratory Data Analysis (EDA)

Age Distribution

Majority of customers fall in **mid-age groups** (20–70 range)

Purchase by Category

Clothing, Footwear, Electronics & Miscellaneous all average **~$40 USD**

Key Observation

Loyal customers contribute significantly higher revenue vs. new users

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NLP — Sentiment Analysis

Process

01

Clean Text

Lowercase, remove stopwords

02

Score Polarity

Sentiment scoring via TextBlob

03

Classify

Positive / Negative / Neutral

~3,800 of 3,900 reviews classified as **Positive**. Positive sentiment customers are more likely to subscribe; negative sentiment highlights product/service issues.

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Feature Engineering & Model

Selected Features

- **Age & Gender**
- **Purchase Amount (USD)**
- **Previous Purchases**
- **Sentiment Score**

Target: Subscription Status (Yes/No)

Random Forest Classifier

Train/Test split: **80/20**

Class	Precision	Recall	F1
No	0.73	0.05	0.08
Yes	0.32	0.86	0.48

Overall Accuracy: 71.2% — model successfully identified key factors influencing subscription behaviour.

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Key Insights

Repeat Customers Subscribe More

Loyalty strongly correlates with subscription uptake

Positive Sentiment = Higher Engagement

Sentiment score is a strong predictor of subscription

Higher Spend → More Subscriptions

Purchase amount increases subscription probability

Category Drives Loyalty

Certain product categories foster stronger customer retention

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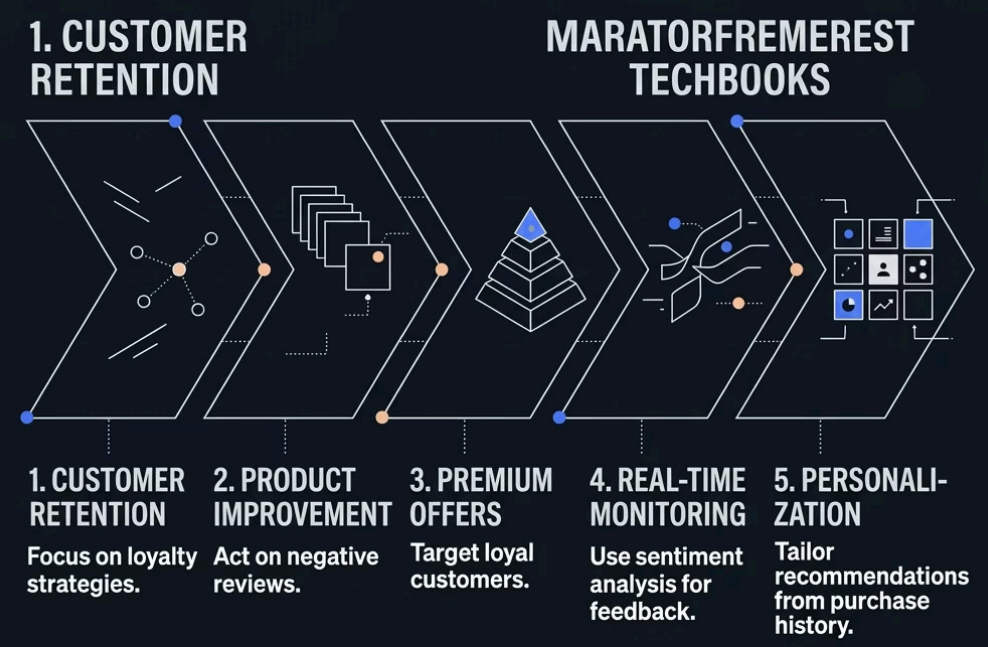
Business Recommendations



Strategic Priorities

→ Retention First

Loyal customers are your highest-value segment

→ Act on Feedback

Negative reviews signal fixable product/service gaps

→ Personalize at Scale

Use purchase history to drive targeted recommendations

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Conclusion

This project demonstrates how **Machine Learning and NLP** can be combined to analyze customer behaviour effectively. By leveraging structured data alongside unstructured review text, businesses can gain deeper insights, improve decision-making, and enhance customer experience.

Limitations

Dataset size, sarcasm in sentiment, and data quality affect model performance

Next Steps

Expand dataset, refine NLP models, and deploy real-time prediction pipeline

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